

Tech & Teach

Digital Chip Design Program

Design RTL

names:

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Task :06

2026\07\13

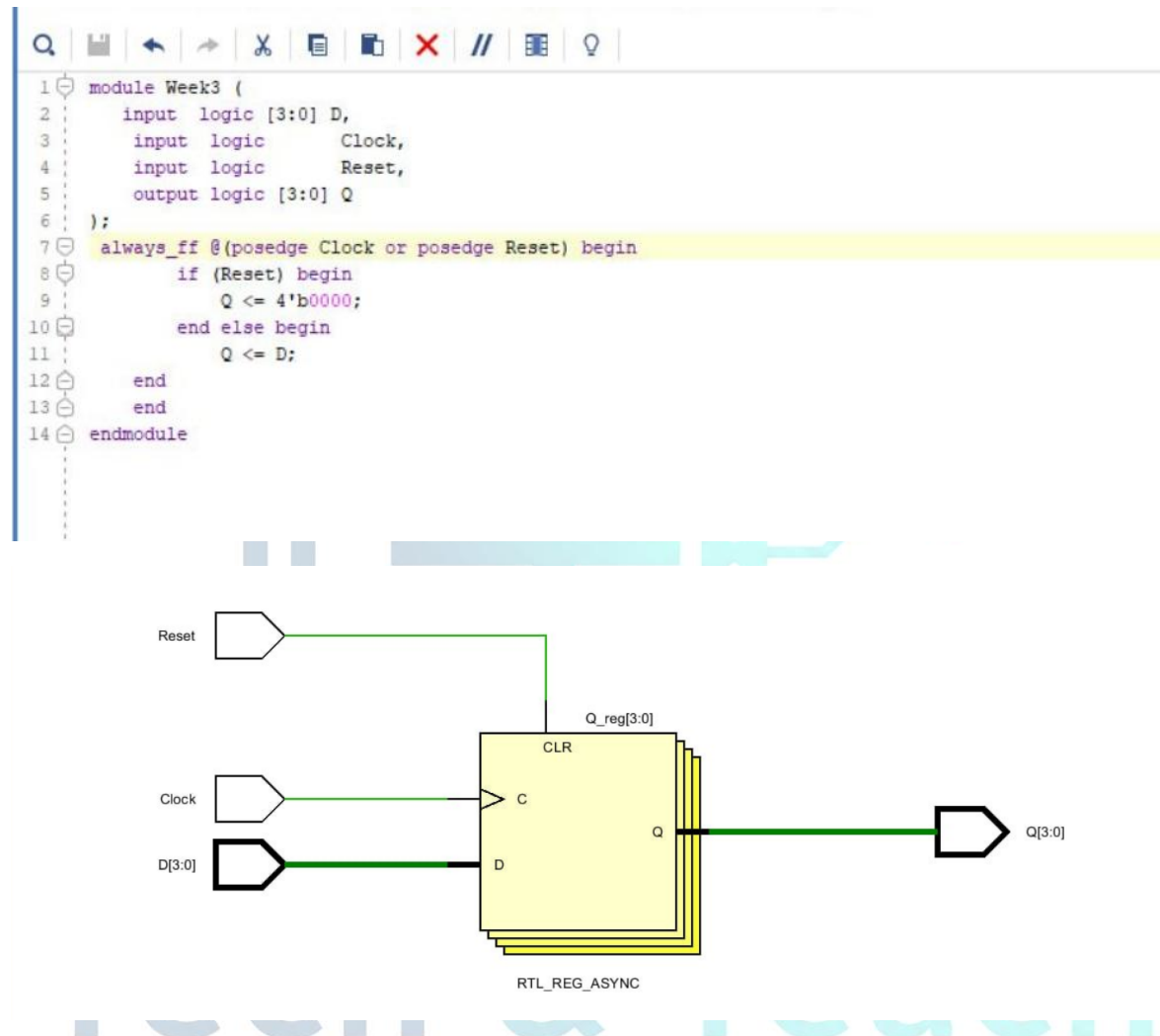
Task topic:

Sequential Datapath Design Part 1 - Blocking & Non-Blocking

Version 1.0

Task: 1.1

1.1.1.1



1.1.2

```

1 module tb_Week3;
2     logic [3:0] D;
3     logic      Clock;
4     logic      Reset;
5     logic [3:0] Q;
6     Week3 uut (
7         .D(D),
8         .Clock(Clock),
9         .Reset(Reset),
10        .Q(Q)
11    );
12    always #5 Clock = ~Clock;
13    initial begin
14        Clock = 0;
15        Reset = 0;
16        D = 4'b0000;
17    end
18    always begin
19        #5 Reset = 1;
20        #10 Reset = 0;
21        #10 D = 4'b1010;
22        #10 D = 4'b0101;
23        #10 D = 4'b1111;
24        #10 D = 4'b0000;
25        #10 D = 4'b1100;
26        #5 Reset = 1;
27        #10 Reset = 0;
28        #10;
29    end
30 endmodule

```



Task: 1.2

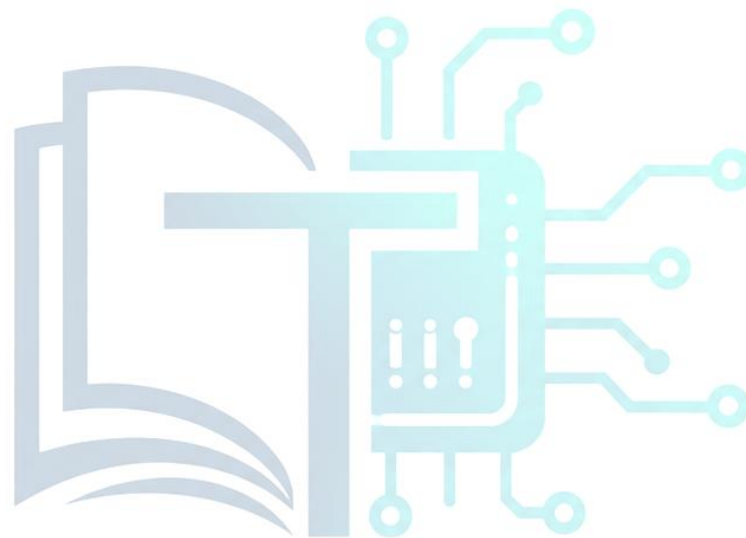
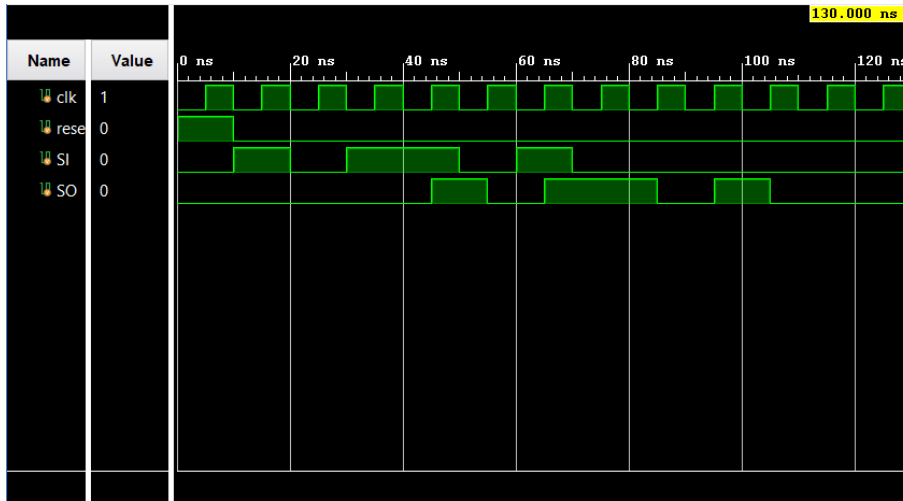
1.2.1.1

```
1 module shift_register(  
2     input logic clk,  
3     input logic reset,  
4     input logic SI,  
5     output logic SO  
6 );  
7 logic D3,D2,D1,D0;  
8 always_ff @(posedge clk or posedge reset)  
9 begin  
10     if(reset)  
11     begin  
12         D3 <= 0;  
13         D2 <= 0;  
14         D1 <= 0;  
15         D0 <= 0;  
16     end  
17     else  
18     begin  
19         D3 <= SI;  
20         D2 <= D3;
```

Test B

```
1 `timescale 1ns/1ps  
2 module shift_register_tb;  
3 logic clk;  
4 logic reset;  
5 logic SI;  
6 logic SO;  
7 shift_register uut(  
8     .clk(clk),  
9     .reset(reset),  
10    .SI(SI),  
11    .SO(SO)  
12 );  
13 // Clock Generation  
14 always #5 clk = ~clk;  
15 initial begin  
16     clk = 0;  
17     reset = 1;  
18     SI = 0;  
19     #10;  
20     reset = 0;
```

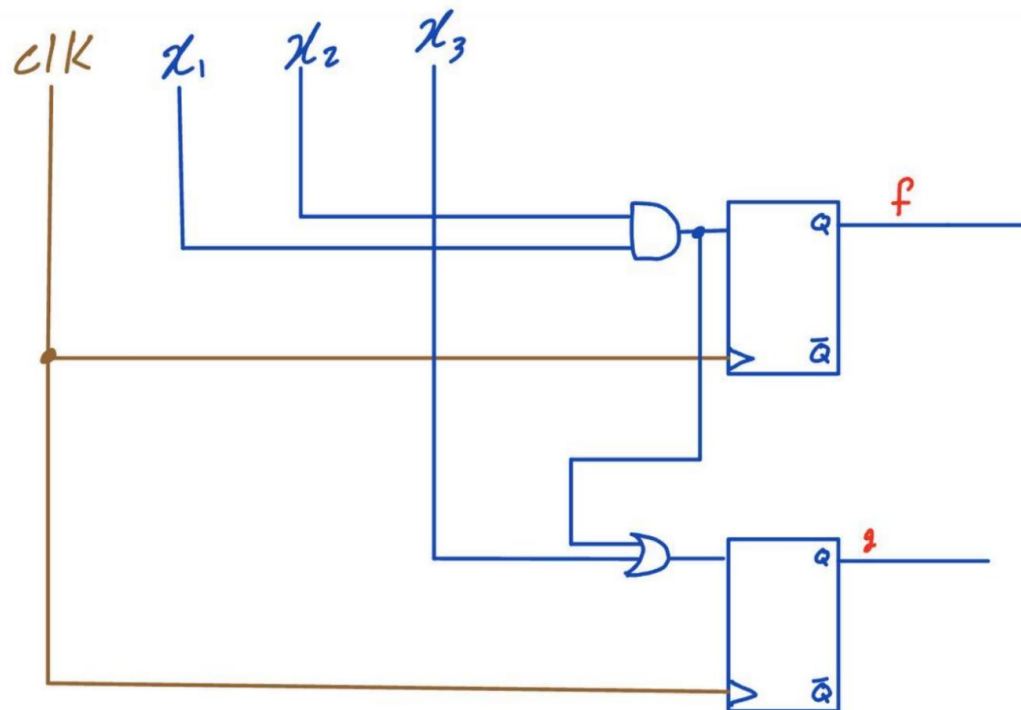
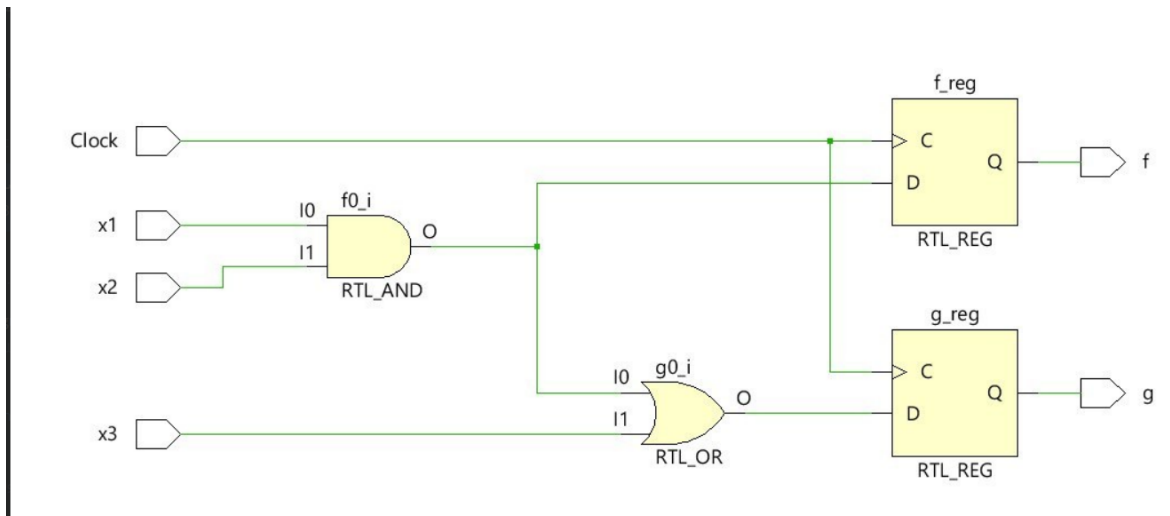
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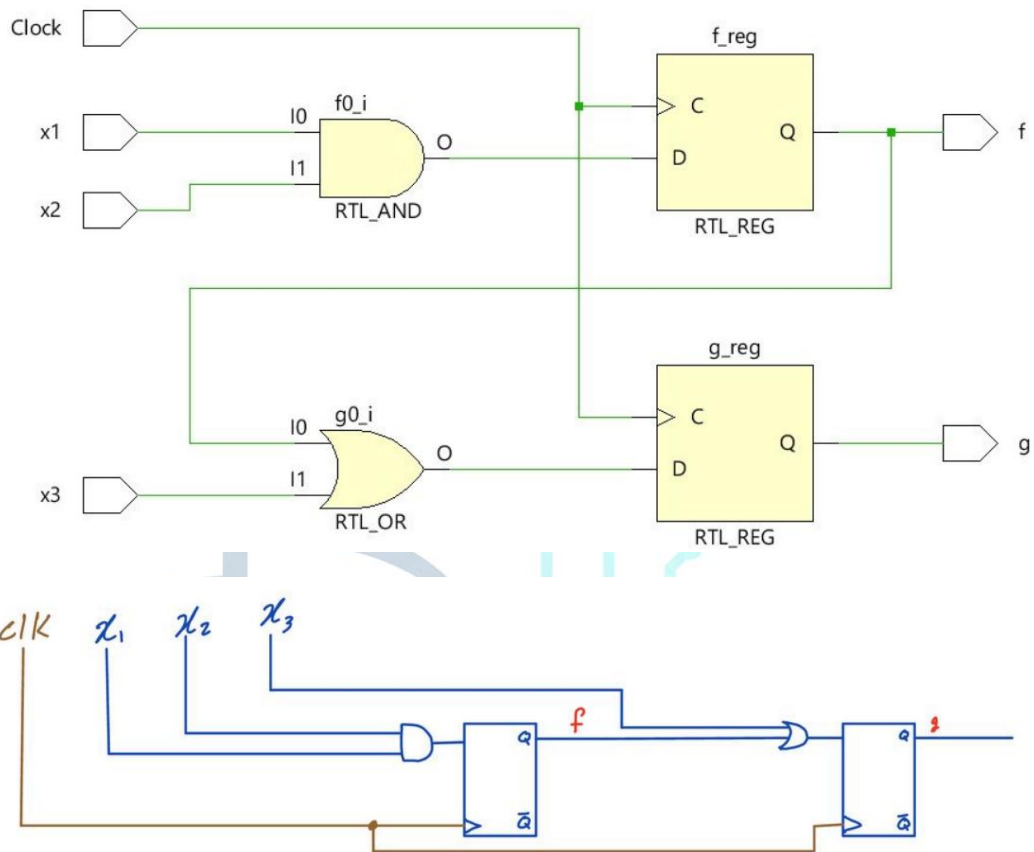
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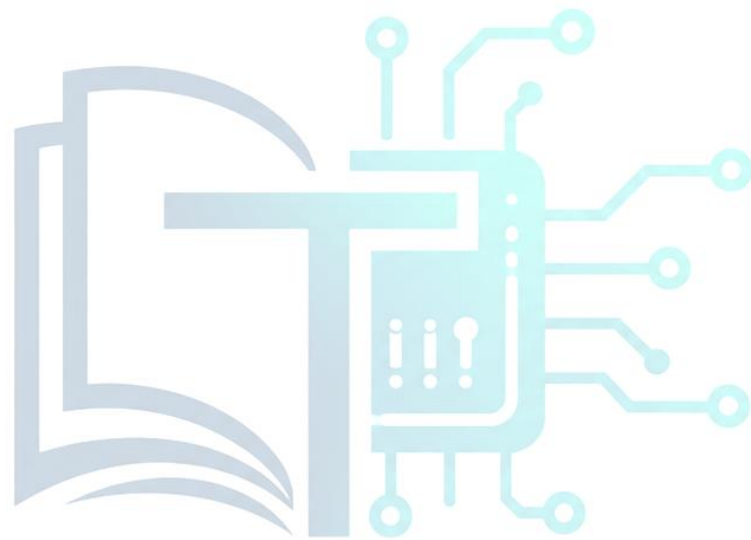
Task: 1.3

1.3.1.1



1.3.1.2





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Work distribution

Section	shaher	Suliman	Abdullah
Report	✓		
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	

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